

Sparse convex hulls for network flows coupled to a simplex

Abstract

We study the convex hull of selected bilinear products between a bounded equality-constrained network flow and a simplex vector. The network and the retained product coordinates provide two distinct sources of structure: circulations factor across undirected blocks, and each block need retain only the simplex states observed there. We develop exact formulations and original-coordinate descriptions from these reductions, and examine the limits of small-coefficient descriptions. The starting point is classical simplex-vertex disaggregation; the structural question is how much of that formulation is needed for a sparse set of products.

1 Introduction

Network models often combine continuous routing decisions with a choice among mutually exclusive services or operating modes. A route-specific adjustment to the cost of a service gives a product of an arc flow and a service weight. Only a small fraction of the possible route-service pairs may occur in the model. Convexifying each product independently discards restrictions linking flows through the network and linking service weights through the simplex. An exact joint hull retains these restrictions, but its standard extended formulation introduces a copy of every arc flow for every simplex vertex.

This paper studies the structure of that joint hull when only selected products are retained. We distinguish the size of an extended formulation from the coefficients and separation cost of an original-coordinate description. We also distinguish equality-constrained flow models from network models with balance inequalities: adding slack arcs to convert the latter to the former can change the graph structure on which a specialized result depends. The elementary foundations below make the state and graph decompositions explicit, including degenerate capacities and zero simplex weights.

2 Model, prior work, and disaggregation

2.1 The sparse network-simplex hull

Let $D = (V, E)$ be a finite directed multigraph. Parallel arcs and self-loops are permitted. Its incidence matrix $A \in \{-1, 0, 1\}^{V \times E}$ has a -1 at the tail and a $+1$ at the head of a non-loop arc; a self-loop has a zero column. Thus $Ax = b$ uses the convention *incoming minus outgoing*. For $b \in \mathbb{Q}^V$ and finite $u \in \mathbb{Q}_+^E$, set

$$P = \{x \in \mathbb{R}^E : Ax = b, 0 \leq x \leq u\}. \quad (1)$$

Lower bounds can be handled by translating x and the corresponding product coordinates. Throughout, graph assumptions concern the graph in (1) after any such modeling transformations.

For an integer $m \geq 0$, write $[m] = \{1, \dots, m\}$ and

$$\Delta_m = \{y \in \mathbb{R}_+^m : \mathbf{1}^\top y \leq 1\}, \quad \lambda_j = y_j \ (j \in [m]), \quad \lambda_0 = 1 - \mathbf{1}^\top y. \quad (2)$$

The *states* are the $d = m + 1$ vertices $e^0 = 0, e^1, \dots, e^m$ of Δ_m ; e^j is the j th unit vector when $j \geq 1$. The residual state is state 0. An observation set $\mathcal{O} \subseteq E \times [m]$ specifies the products retained as original coordinates. We study

$$S = \{(x, y, z) : x \in P, y \in \Delta_m, z_{ej} = x_e y_j \quad ((e, j) \in \mathcal{O})\}, \quad (3)$$

$$H = \text{conv } S. \quad (4)$$

An unobserved product is absent from z , even if it is introduced temporarily in an extended formulation. State 0 has no observed coordinates. We use $\text{conv } \emptyset = \emptyset$. When $m = 0$, y and z are empty vectors and $H = P$.

The original coordinates are (x, y, z) . An extended formulation is a linear system in these coordinates and additional continuous variables whose projection is H . A separation procedure must either certify membership or produce a linear inequality valid for all of H and strictly violated by the candidate. Unless specified otherwise, a facet is taken relative to the affine hull. A bound on integer coefficients presupposes a stated normalization; a ratio of two nonzero coefficients is invariant under scaling the inequality. Arithmetic-operation bounds are distinguished from bounds on rational encoding length.

For comparison, the usual independent product relaxation adds, for every $(e, j) \in \mathcal{O}$, the four inequalities

$$0 \leq z_{ej}, \quad z_{ej} \leq u_e y_j, \quad z_{ej} \leq x_e, \quad z_{ej} \geq x_e - u_e(1 - y_j) \quad (5)$$

to $x \in P$ and $y \in \Delta_m$. These are the McCormick inequalities for the box $[0, u_e] \times [0, 1]$ [11]. They need not describe the joint hull in (4).

2.2 Relation to prior work

The general convexification principle is established. Disjunctive hull formulations [3] apply after expanding the simplex into its vertices. Simultaneous convexification over a polytope times a simplex is developed by Davarnia et al. [5]. Khademnia and Davarnia [8] specialize this setting to network polytopes: their Appendix, equation (25), gives a full extended hull, and their aggregation-and-projection framework gives a complete original-space description in principle. Their explicit tree and forest constructions identify more structured inequality families; the forest recipe for multiple simplex coordinates should not be confused with the entire projection framework. Their model permits balance inequalities, whereas (1) imposes equalities. We use the published version for equation and example numbering.

The shared-simplex gluing principle is also known. In particular, Davarnia [4, Proposition 2.6] proves that convexifying Cartesian components separately is exact when the components share the same simplex. The cited locator belongs to the dissertation, which is distinct from the 2017 journal article. Our block decomposition applies this principle to the independent circulation spaces of an equality-constrained network. We give a direct proof to make the treatment of sparse observations and zero weights explicit.

Several established methods explain the subsequent structural reductions. Reduced reformulation–linearization uses multiplied linear equations to remove common-factor products [10]. In networks, the remaining freedom can be identified with circulations supported on unobserved arcs. The network representation of Cayley hulls of Kis and Horváth [9] gives cut descriptions and transportation projections; their Section 5.7 is a close predecessor for the parallel-path construction. The support geometry of Minkowski sums and edge-generated zonotope refinements is classical [7, 12]. These are ingredients of the graph-specific formulations and separation procedures, rather than new general convexification mechanisms.

Two further comparisons delimit coefficient obstructions. Three-way transportation universality is established by De Loera and Onn [6], and large coefficients in general 0/1 polytopes are classical [2]. A restricted network realization must therefore be specified as part of a coefficient result. Moreover, the integrality of *aggregate* flows on two-terminal series-parallel networks does not give a hull description in selected state-flow coordinates. Almoghrabi et al. [1, Remark 1] explicitly distinguish aggregate integrality from decomposability of the vector of individual commodity flows. Our observed products retain such individual-state information. Neither a sparse extended formulation nor polynomial separation precludes large numerical coefficients after projection.

2.3 The known exact disaggregation

For $\lambda \geq 0$, define the scaled flow system

$$P(\lambda) = \{f \in \mathbb{R}^E : Af = \lambda b, 0 \leq f \leq \lambda u\}. \quad (6)$$

For $\lambda > 0$, this is λP ; at $\lambda = 0$ it is $\{0\}$, including when P is empty. Using (6) avoids assigning an ambiguous meaning to $0\emptyset$.

Proposition 2.1 (Simplex-vertex disaggregation). *A point (x, y, z) belongs to H if and only if $y \in \Delta_m$ and there exist state flows $f^0, \dots, f^m \in \mathbb{R}^E$ such that*

$$f^k \in P(\lambda_k) \quad (k \in \{0, \dots, m\}), \quad \sum_{k=0}^m f^k = x, \quad f_e^j = z_{ej} \quad ((e, j) \in \mathcal{O}). \quad (7)$$

The equivalence holds even if P is empty. If the system is feasible, it provides a convex decomposition into at most $m + 1$ points of S .

Proof. For a point of S , set $f^k = \lambda_k x$. This satisfies all displayed relations, since $\sum_k \lambda_k = 1$. The system is linear in (x, y, z, f) , so its projection is convex and contains H .

Conversely, suppose the system is feasible. If $\lambda_k = 0$, the finite upper bounds give $f^k = 0$. For each positive weight let $x^k = f^k / \lambda_k \in P$ and define the graph point $p^k = (x^k, e^k, z^k)$ by $z_{ej}^k = x_e^k (e^k)_j$. Its weighted average has x coordinate $\sum_k f^k = x$, y coordinate $\sum_k \lambda_k e^k = y$, and observed product coordinate $\lambda_j x_e^j = f_e^j = z_{ej}$ when $\lambda_j > 0$. If $\lambda_j = 0$, that observed coordinate is zero on both sides. Thus $(x, y, z) = \sum_{k: \lambda_k > 0} \lambda_k p^k \in H$. There is at least one positive weight, so feasibility also implies $P \neq \emptyset$. This proves the empty-set assertion. $\square$

The formulation has $(m + 1)|E|$ state-flow variables before elimination. Its rational size is polynomial in the ordinary network and simplex input, so generic polynomial-time hull optimization and separation already follow from linear programming. The structural issue is avoiding this full state-by-arc expansion, or projecting it into useful original-coordinate inequalities. In particular, eliminating $f^0 = x - \sum_{j=1}^m f^j$ keeps the residual bounds

$$0 \leq x - \sum_{j=1}^m f^j \leq \left(1 - \sum_{j=1}^m y_j\right)u. \quad (8)$$

These bounds cannot in general be discarded.

3 Independent blocks and locally merged states

3.1 Circulation factorization

All graph connectivity terminology below refers to the undirected multigraph underlying D . Isolated vertices impose only their balance equation. Let E_{br} be the bridges. Let $\mathcal{B}$ be the cyclic blocks: the maximal biconnected edge blocks that contain a cycle, with each self-loop treated as its own block. A pair of parallel edges can form a cycle of length two. The sets E_{br} and E_B for $B \in \mathcal{B}$ partition E . The cycle rank of a connected block is

$$r_B = |E_B| - |V_B| + 1. \quad (9)$$

In particular, a loop has rank one. Rank refers to the ambient circulation space, even when capacities reduce the dimension of the feasible polytope.

The equation $Av = b$ is solvable exactly when the entries of b sum to zero on every connected component. Necessity follows by summing the incidence rows in a component. For sufficiency, choose a spanning forest, set the nonforest coordinates to zero, and determine each tree-edge flow by the balance sum on one side of its fundamental cut, with the sign determined by the arc orientation. This satisfies every balance equation. This construction uses $O(|V| + |E|)$ arithmetic operations, and for rational b produces a rational v of polynomial encoding length. The reference flow v need not obey its capacity bounds. If the component condition fails, P and H are empty.

Fix such a reference vector v . For each B , let A_B be the incidence matrix restricted to columns in E_B and rows in V_B , and define

$$K_B = \{h \in \mathbb{R}^{E_B} : A_B h = 0, -v_{E_B} \leq h \leq u_{E_B} - v_{E_B}\}. \quad (10)$$

Unlike P , this is a domain of circulation *deviations*; it need not contain the origin.

Lemma 3.1 (Block factorization). *Every $g \in \ker A$ vanishes on E_{br} and restricts to a circulation on each cyclic block. Conversely, combining arbitrary block circulations and zero bridge coordinates gives an element of $\ker A$. Consequently, if every bridge satisfies $0 \leq v_e \leq u_e$, the map*

$$x_e = v_e \quad (e \in E_{\text{br}}), \quad x_{E_B} = v_{E_B} + h_B \quad (B \in \mathcal{B}) \quad (11)$$

is an affine bijection from $\prod_{B \in \mathcal{B}} K_B$ to P . If a bridge bound fails, P is empty. The product of an empty family of block domains is a singleton.

Proof. Choose a spanning forest of the underlying multigraph, omitting loops. Each nonforest edge gives a signed fundamental-cycle circulation: use the unique forest path between its ends, or just the edge itself for a loop. These vectors form a basis of $\ker A$. Indeed, subtracting from a circulation the combination that cancels all nonforest coordinates leaves a circulation supported on a forest, which is zero by successive leaf elimination.

Each fundamental cycle lies in a single cyclic block. Hence its vector is zero on every bridge, and on every block it is either zero or a circulation of that block. The basis expansion proves the first assertion. For the converse, extend each block circulation by zero to E . Its incidence is zero at every vertex, including articulation vertices, so their sum is a global circulation. The sum is direct because the block edge sets are disjoint. Applying this decomposition to $x - v$ proves that $Ax = b$ is equivalent to the circulation conditions and the fixed bridge coordinates in (11). The bounds then separate by edge block, giving precisely (10) and the bridge conditions. $\square$

In particular, an articulation vertex does not carry an additional coupling variable between block deviations. Its fixed net balance is already carried by v ; every block deviation has zero net balance at that vertex separately. Any spanning tree in B gives a fundamental-cycle matrix $C_B \in \{0, \pm 1\}^{E_B \times r_B}$ with chord rows equal to the identity. Thus $h_B = C_B t_B$ gives unique cycle coordinates $t_B \in \mathbb{R}^{r_B}$. These coordinates remain useful when K_B is lower-dimensional.

3.2 The state merger and its exact refinement

For a nonempty convex set K , use the usual scalar multiplication $\alpha K = \{\alpha h : h \in K\}$, so $0K = \{0\}$. For nonnegative scalars,

$$\alpha_1 K + \cdots + \alpha_q K = (\alpha_1 + \cdots + \alpha_q) K. \quad (12)$$

If the total weight is positive, the inclusion from left to right follows by dividing by that weight and taking a convex combination; the reverse inclusion uses the same point of K in each summand. If the total is zero, both sides are $\{0\}$. An empty sum is also interpreted as $\{0\}$.

For each block define its observed labels and their count by

$$J_B = \{j \in [m] : (e, j) \in \mathcal{O} \text{ for some } e \in E_B\}, \quad a_B = |J_B|. \quad (13)$$

This is a structural definition: a label in J_B remains present when $y_j = 0$. Let $*_B$ denote a local merged state, distinct from the global residual state 0, and set

$$I_B = J_B \cup \{*_B\}, \quad \lambda_{B,j} = y_j \ (j \in J_B), \quad \lambda_{B,*} = 1 - \sum_{j \in J_B} y_j. \quad (14)$$

The merged state collects all labels absent from J_B and state 0. Different blocks may merge different global states.

Theorem 3.2 (Exact blockwise state reduction). *Assume $P \neq \emptyset$. A point (x, y, z) belongs to H if and only if $x \in P$, $y \in \Delta_m$, every observed bridge satisfies*

$$z_{ej} = v_e y_j \quad (e \in E_{\text{br}}, (e, j) \in \mathcal{O}), \quad (15)$$

and for every cyclic block B there exist vectors $h^{B,k} \in \mathbb{R}^{E_B}$, $k \in I_B$, satisfying

$$h^{B,k} \in \lambda_{B,k} K_B \quad (k \in I_B), \quad (16)$$

$$\sum_{k \in I_B} h^{B,k} = x_{E_B} - v_{E_B}, \quad (17)$$

$$h_e^{B,j} = z_{ej} - y_j v_e \quad ((e, j) \in \mathcal{O}, e \in E_B). \quad (18)$$

The block conditions yield state flows in (7), and hence a decomposition into at most $m + 1$ graph points, by explicit proportional refinement.

Proof. By lemma 3.1, every K_B is nonempty. Start with state flows in (7). Their deviations $g^k = f^k - \lambda_k v$ are circulations. On B , they satisfy $g_{E_B}^k \in \lambda_k K_B$: division proves this for positive weights, and both deviations and scaled domains are zero for zero weights. For $j \in J_B$ take $h^{B,j} = g_{E_B}^j$, and sum the other deviations into $h^{B,*}$. Identity (12) proves (16). Summing the deviations proves (17), and the observed coordinates give (18). On bridges the deviations vanish, proving (15).

Conversely, fix feasible local vectors. For each global state k and block B , if $k \in J_B$, set $g_{E_B}^k = h^{B,k}$. For the remaining states use

$$g_{E_B}^k = \begin{cases} (\lambda_k/\lambda_{B,*})h^{B,*}, & \lambda_{B,*} > 0, \\ 0, & \lambda_{B,*} = 0. \end{cases} \quad (19)$$

If the merged weight is zero, every constituent weight is zero as well. In every case $g_{E_B}^k \in \lambda_k K_B$, and the refined vectors sum to the block deviation. Put $g_e^k = 0$ on bridges and combine blocks. The block factorization gives $Ag^k = 0$. Then $f^k = \lambda_k v + g^k$ satisfies $Af^k = \lambda_k b$. Its capacity bounds hold on blocks by the scaled domains, and on bridges because $P \neq \emptyset$ implies $0 \leq v_e \leq u_e$. Equations (17) and (18), together with the bridge equations, give $\sum_k f^k = x$ and all observed products. Apply [proposition 2.1](#). $\square$

The domain condition (16) is itself linear:

$$A_B h^{B,k} = 0, \quad -\lambda_{B,k} v_{E_B} \leq h^{B,k} \leq \lambda_{B,k} (u_{E_B} - v_{E_B}). \quad (20)$$

Finite bounds force $h^{B,k} = 0$ at zero weight. No division by a variable is needed in the formulation; division appears only in the recovery proof at strictly positive weights. If $J_B = \emptyset$, the block has one merged state of weight one and its conditions reduce to $x_{E_B} - v_{E_B} \in K_B$, already enforced by $x \in P$. Thus unobserved blocks require no new variables or constraints.

The theorem is a constructive specialization of the known shared-simplex gluing principle. It does not introduce independent simplex vectors for different blocks: all use the same global y . It also does not merge observed state slices, whose domains can differ after the products are fixed.

Example 3.3 (A common matrix is not sufficient for merging). Let $M = (-1, 1, 2)^\top$, $c^1 = (0, 1, 100)^\top$, and $c^2 = (0, 100, 2)^\top$. Both polytopes $Q_i = \{q \in \mathbb{R} : Mq \leq c^i\}$ equal $[0, 1]$. For weights $\alpha_1 = \alpha_2 = 1/2$, the weighted Minkowski sum is still $[0, 1]$, whereas the aggregated rows $Mq \leq (c^1 + c^2)/2$ allow $q = 2$. The two upper bounds become $q \leq 101/2$ and $2q \leq 51$. Thus replacing state-specific systems by the sum of their right-hand sides is generally inexact, even for a common matrix and nonempty bounded domains. This is the standard reaggregation boundary discussed by Kis and Horváth [9, Section 2, equation (7)]. Identity (12) uses the stronger fact that the domains being merged are scaled copies of the same convex set.

Finally, these results concern the hull of (3). Intersecting H with additional linear side constraints yields a valid relaxation of the graph with those constraints, but need not give its exact convex hull. Such constraints can couple blocks or alter conditional state domains; exactness then requires a separate argument.

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